

BUGFIXES

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Root-cause-analysed bug fixes for this project, per the Universal Mandatory Constraints (CLAUDE.md mandate #10). Each entry records the defect, root cause, affected files, the fix, and a link to the reproduction/verification evidence.

BUGFIX-0001 — run-tests.sh aborts after the first test under set -e

- **Type:** Bug (script-internal failure — Helix Constitution §11.4.1)
- **Status:** Fixed
- **Date:** 2026-06-30
- **Affected file:** tests/run-tests.sh (test_result())

Symptom

bash tests/run-tests.sh printed only its banner and the first test's section header, then exited 1 (7 lines of output total) — every test after the first was silently never run, and no TEST SUMMARY was printed.

Reproduction (captured, run in-session before the fix)

```
$ bash -c 'set -euo pipefail; x=0; ((x++)); echo "SURVIVED,
x=$x"; echo "exit=$?"
exit=1                                # "SURVIVED" never printed

$ bash tests/run-tests.sh 2>&1 | wc -l
7                                     # aborted; exit=1
```

Root cause (FACT — not a guess, Helix Constitution §11.4.6)

`test_result()` counted with the bash post-increment idiom `(( TESTS_RUN++ ))`. The arithmetic command `(( expr ))` returns exit status **1** when `expr` evaluates to **0**. `TESTS_RUN++` is a *post*-increment, so its value is the value *before* incrementing — 0 on the very first call — making `(( TESTS_RUN++ ))` return status 1. The script runs under `set -euo pipefail` (line 7), so that non-zero status aborted the entire suite at the first `test_result` call, before any results or the summary could print. The same trap applied to the first PASS `(( TESTS_PASSED++ ))` and first FAIL `(( TESTS_FAILED++ ))`.

This was a **pre-existing** latent defect (the `(( ... ++ ))` idiom was original); it surfaced while wiring the constitution-inheritance pre-flight gate in as the first test.

Fix (at source, Helix Constitution §11.4.1)

Replaced the three post-increment arithmetic commands with the assignment form, which always returns status 0 and is immune to the `set -e` trap:

```
- ((TESTS_RUN++))
+ TESTS_RUN=$((TESTS_RUN + 1))
- ((TESTS_PASSED++))
+ TESTS_PASSED=$((TESTS_PASSED + 1))
- ((TESTS_FAILED++))
+ TESTS_FAILED=$((TESTS_FAILED + 1))
```

Verification (captured, run in-session after the fix)

```
$ bash -c 'set -euo pipefail; x=0; x=$((x+1)); echo "SURVIVED,
x=$x"; echo "exit=$?"
SURVIVED, x=1
exit=0
```

```
$ bash tests/run-tests.sh 2>&1 | wc -l
28                                # suite now runs to completion (was 7)
# first line of results:
✓ PASS: Constitution inheritance (§11.4.35)
```

The suite now executes every test and prints its summary. (Its overall exit remains non-zero in an environment without the running proxy services — those are real-infrastructure tests that correctly FAIL/skip when the live System is absent, Helix Constitution §11.4.11 — that is a separate, pre-existing, infrastructure-dependent condition, not this defect.)

Cross-reference (§11.4.138): the "proxy services not running" condition noted above was treated as merely environmental. It was not — BUGFIX-0002 below is the *reason* the proxy would not stay up under rootless Podman. Driving the real data path (not just "tests pass") surfaced it.

BUGFIX-0002 — squid crash-loops under rootless Podman (log dir not writable) → proxy never serves

- **Type:** Bug (product defect — end-user-visible: the proxy does not work)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected files:** lib/container-runtime.sh (create_directories()), start (init_cache())
- **Regression guard:** tests/regression/log_dir_writable_test.sh (§11.4.135, §11.4.115 RED_MODE polarity)
- **Workable item:** #38 (the live-usability proof stream)

Symptom

A freshly-booted proxy (./start --no-vpn, rootless Podman) returned http_code=000 for every request through it. proxy-squid was not (healthy) — it was restarting in a loop, never accepting a connection, so **no feature of the proxy worked for the end user** — the exact "tests pass but the feature can't be used" failure mode §11.4 forbids.

Reproduction (captured, run in-session before the fix — §11.4.115 RED)

```
# Live boot of the real orchestrator, then curl through the proxy:
$ curl -s -o /dev/null -w '%{http_code}' -x http://127.0.0.1:53128
http://example.com/
000
```

```
# squid's own log (read from inside the container) showed the
FATAL:
$ podman logs proxy-squid | tail
FATAL: Cannot open '/var/log/squid/access.log' for writing.
       The parent directory must be writeable by the user 'proxy',
       which is the cache_effective_user set in squid.conf.
```

```
# Unit reproduction of the orchestrator code path (pre-fix
replica):
$ RED_MODE=1 tests/regression/log_dir_writable_test.sh
```

```
[PASS] BUGFIX#38 log-dir-writable (RED_MODE=1): RED reproduced:  
pre-fix LOG_DIR mode=755 is NOT world-writable
```

Root cause (FACT — not a guess, §11.4.6)

Under **rootless** Podman the invoking host uid (1000) maps to the container's root, but every other container uid is remapped through `/etc/subuid` into a high host range. `squid.conf` sets `cache_effective_user proxy`, so squid drops privileges to its non-root proxy user — which, on the host, is a high subuid that owns **none** of the host-created bind-mount dirs. The host `./logs` directory was created mode `0755` (owner-write only), so the container proxy user had only world `r-x` — no write — and squid `FATALed` opening `access.log`, then crash-looped. The orchestrator already `chmod 777`-ed the **cache** bind-mount (`init_cache`) for this exact reason, but the **log** bind-mount was missed — a single missing site of an already-known remedy.

Fix (at source, §11.4.1 / §11.4.108 SOURCE layer)

Make the squid log bind-mount world-writable in the orchestrator's `dir-init`, mirroring the existing `cache chmod 777`. Applied at **both** self-sufficient sites so it holds regardless of call order:

```
# lib/container-runtime.sh  create_directories()  
    mkdir -p "$LOG_DIR"  
+ chmod 777 "$LOG_DIR"  
  
# start  init_cache()  
+ mkdir -p "$LOG_DIR"  
+ chmod 777 "$LOG_DIR"
```

Verification (captured, run in-session after the fix)

Unit — regression guard, §11.4.115 polarity (RED reproduces, GREEN proves):

```
$ RED_MODE=1 tests/regression/log_dir_writable_test.sh  # pre-fix  
replica  
[PASS] RED reproduced: LOG_DIR mode=755 NOT world-writable  
$ tests/regression/log_dir_writable_test.sh             # real  
fixed code  
[PASS] GREEN: create_directories() made LOG_DIR mode=777 world-  
writable
```

§1.1 paired mutation (guard is not a tautology): stripping the `chmod 777 "$LOG_DIR"` from `create_directories()` made the GREEN guard **FAIL** (REGRESSION: ... mode=755 NOT world-writable, exit 1); the

lib was restored byte-identical (md5
0128a96b6d467c2da5b7cef8a808e563 before == after) and the guard
PASSED again.

Live data-plane (§11.4.108 RUNTIME + USER-VISIBLE):

after the fix the booted proxy serves real HTTP through it:

```
$ curl -x http://127.0.0.1:53128 http://example.com/ ->  
http_code=200  
Via: 1.1 proxy-squid (squid/6.13)  
# squid access.log (the file that FATAL'd pre-fix) – real request  
logged:  
1782858066.886 189 10.89.1.249 TCP_MISS/200 951 GET http://  
example.com/ - HIER_DIRECT/104.20.23.154 text/html  
$ podman ps -> proxy-squid Up (healthy)
```

Evidence: qa-results/regression/bugfix38/.

BUGFIX-0003 — tests/run-tests.sh aborts mid-suite on a no-message FAIL (set -e); ~29 of 39 tests silently never ran

- **Type:** Bug (script-internal failure — §11.4.1; sibling of BUGFIX-0001)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected file:** tests/run-tests.sh (test_result())
- **Regression guard:** tests/regression/
test_result_returns_zero_test.sh (§11.4.135, §11.4.115 RED_MODE
polarity)

Symptom

bash tests/run-tests.sh printed only ~10 results then stopped —
no TEST SUMMARY, exit 1. The Directory / Config / Compose /
Runtime / Port / Cache / VPN / Service-startup groups after the
first FAIL never ran, so the suite *appeared* to run while actually
exercising under a third of its tests and hiding real failures.

Reproduction (captured, run in-session before the fix — §11.4.115 RED)

```
# Extract the pre-fix test_result and call it with a no-message  
FAIL under set -e:  
$ bash -c 'set -euo pipefail; <test_result>; test_result "x"  
"FAIL"; echo SURVIVED'  
x FAIL: x
```

```
# exit=1 – "SURVIVED" never printed (the function returned 1 and
aborted)
```

```
$ RED_MODE=1 tests/regression/test_result_returns_zero_test.sh
[PASS] RED reproduced: pre-fix test_result aborts under set -e on
a no-message FAIL
```

Root cause (FACT — not a guess, §11.4.6)

test_result()'s FAIL branch ended on `[[ -n "$message" ]] && echo -e " → $message"`. When message is empty (every test_result "... " "FAIL" call with no third argument), the `[[ -n "$message" ]]` test is false, the `&&` list short-circuits, and the list — the **last command of the function** — returns exit status **1**. When such a test_result is the last command executed in a test function (e.g. the final "Upstreams" iteration of test_directories's for loop, where the dir is absent → FAIL), that function returns 1, and under set -e pipefail main() aborts immediately — every later test group and the summary never run. This is the same §11.4.1 class as BUGFIX-0001 (a reporting helper leaking a non-zero status into control flow), a *different* site (test_result's tail, not the ((++)) counter), so BUGFIX-0001's fix did not cover it.

Fix (at source, §11.4.1)

test_result is a reporting helper; its exit status must never gate control flow. Append an explicit return 0:

```
        [[ -n "$message" ]] && echo -e "  ${YELLOW}→ $message$
        {NC}"
    fi
+
+   return 0
+ }
```

Verification (captured, run in-session after the fix)

```
# §11.4.115 GREEN (real fixed test_result survives a no-message
FAIL):
```

```
$ tests/regression/test_result_returns_zero_test.sh
[PASS] GREEN: real test_result returns 0 on a no-message FAIL
```

```
# Full suite now runs to completion (was 32 lines, aborted):
```

```
$ bash tests/run-tests.sh # 82 lines, reaches TEST SUMMARY
Tests Run:    41
Tests Passed: 34
Tests Failed: 7
```

§1.1 paired mutation (guard is not a tautology): deleting the return 0 re-introduced the abort → the GREEN guard **FAILED** (REGRESSION: test_result aborts the suite ..., exit 1); tests/run-tests.sh was restored byte-identical (md5 7c2bab18c4566d081b1c8aa7a9a412e0 before == after) and the guard PASSED again.

Follow-up (separate, tracked): with the abort fixed, the suite now honestly reports 7 **pre-existing FAILs** (e.g. Directory Upstreams) that the abort had been masking. These are real existing-system findings to triage — not regressions from this fix, but newly *visible* because the suite finally runs to completion.

Evidence: qa-results/regression/bugfix0003/.

BUGFIX-0004 — run-tests.sh reports FAILURE on a HEALTHY serving proxy (§11.4.1 false-FAIL); skips recorded as PASS

- **Type:** Bug (anti-bluff — §11.4.1 false-FAIL + §11.4.3 PASS-by-default-for-skip)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected file:** tests/run-tests.sh (test_result, test_environment, test_directories, test_container_runtime, test_ports + new port_verdict/ports_check_one, test_vpn, test_service_startup, print_summary)
- **Regression guard:** tests/regression/port_topology_aware_test.sh (§11.4.135, §11.4.115 RED_MODE polarity)
- **Bluff audit:** docs/research/existing_test_bluffs_audit/README.md (B7)
- **Workable item:** #47 (P8)

Symptom

Once BUGFIX-0003 let the suite run to completion, bash tests/run-tests.sh reported Tests Run: 41, Passed: 34, Failed: 7 and exited **1 against a perfectly healthy, serving proxy** (proxy-squid + proxy-dante Up (healthy)). A suite that declares FAILURE on a working System is a §11.4.1 false-FAIL — as misleading as a false-PASS.

Reproduction (captured, run in-session before the fix — §11.4.115 RED)

```
# Against the running, healthy proxy:
$ bash tests/run-tests.sh | tail -4
Tests Run:      41
Tests Passed: 34
Tests Failed: 7      # exit 1
# The 7 FAILs: .env file exists, HTTP_PROXY_PORT set, Directory
Upstreams,
# Docker installed, Port 53128 available, Port 51080 available,
Port 58080 available

$ RED_MODE=1 tests/regression/port_topology_aware_test.sh
[PASS] RED reproduced: pre-fix logic classifies a healthy serving
proxy port
      (owner up + listening) as FAIL
```

Root cause (FACT — not a guess, §11.4.6)

Three independent bluff classes, all FACT-confirmed against the live host:

1. **Port-semantics inversion (§11.4.1).** `test_ports` reported any port that was IN USE as FAIL ("Port in use"). The running proxy's own listening ports (squid 53128, dante 51080) are *in use by their own healthy service* — the serving state — so the suite FAILED on health.
2. **Inverted runtime check (§11.4.161).** `test_container_runtime` asserted Docker installed and FAILED on its absence. The project MANDATES rootless Podman and FORBIDS Docker as a workflow — the check tested for the forbidden runtime and failed on the compliant state.
3. **Config-absent-as-FAIL (§11.4.3).** `.env` (gitignored, §11.4.10/ §11.4.30) and `Upstreams/` absence FAILED, though their absence is the *expected* fresh-checkout topology. And (B7) `test_result` had no SKIP state, so `test_vpn/test_service_startup` skipped paths were recorded as PASS, inflating `TESTS_PASSED` (§11.4.3 PASS-by-default).

Fix (at source, §11.4.1 / §11.4.3 / §11.4.161)

- **3-state test_result** — add a SKIP branch + `TESTS_SKIPPED` counter + Tests Skipped: summary line; the suite exit gates on `TESTS_FAILED -eq 0` only (skips are neither pass nor fail). BUGFIX-0001 assignment-form counters + BUGFIX-0003 return 0 preserved.
- **§11.4.3 topology-aware ports** — a *pure* `port_verdict(owner_serving, listening)` truth-table (serving→PASS, owner-up-but-not-listening→FAIL, pre-start-

free→PASS, foreign-process-holds-it→SKIP); _ports_check_one resolves owner_serving from real podman ps/podman port and listening from real ss (no data-plane probe).

- **§11.4.161** — assert the mandated podman; Docker informational only, absence is SKIP not FAIL.
- **§11.4.3 SKIP** for absent gitignored .env/HTTP_PROXY_PORT (validate the tracked .env.example template instead) and absent upstreams//Upstreams/.
- **B7** — test_vpn/test_service_startup disabled paths emit SKIP.

Verification (captured, run in-session after the fix)

Full suite vs the healthy running proxy – zero failures, honest skips:

```
$ bash tests/run-tests.sh | tail -5
```

```
Tests Run:      43
```

```
Tests Passed:   37
```

```
Tests Skipped:  6
```

```
Tests Failed:   0          # exit 0 – "All tests passed (6 skipped ...)"
```

§11.4.115 polarity guard (tests the REAL pure port_verdict, not a copy):

```
$ RED_MODE=1 tests/regression/port_topology_aware_test.sh  # reproduces
```

```
[PASS] RED reproduced: healthy serving port classified FAIL
```

```
$ tests/regression/port_topology_aware_test.sh              # GREEN
```

```
[PASS] GREEN: HEALTHY=PASS NOTSERVING=FAIL PRESTART_FREE=PASS  
PRESTART_BUSY=SKIP
```

§1.1 paired mutation (guard is not a tautology): flipping port_verdict's PASS/FAIL branch made the GREEN guard **FAIL** (REGRESSION: ... HEALTHY=FAIL ..., exit 1) with bash -n still clean (assertion, not parse error); tests/run-tests.sh restored byte-identical (md5 2f4d3e4bf33cd391036cee595839d439) and the guard PASSED again. The guard is wired into test_regression_guards() (GREEN + RED self-check); the 5 pre-existing guards still PASS.

Honest note (§11.4.6 / §11.4.174): port 58080 (the control-API port) is currently held by a **non-project** host process — no proxy container publishes it — so the suite classifies it SKIP (readiness not assertable), never a false-FAIL blaming the proxy. The foreign process was **not** touched (shared-host ownership). This is a real condition to free before P10 deploys the control-API there.

Evidence: qa-results/p8/, qa-results/regression/portstopology/.

BUGFIX-0005 — final-verify.sh + verify-proxy.sh greened a NO-VPN config (false-VPN-routing §15 bluff) + aborted under set -e

- **Type:** Bug (anti-bluff — false-VPN-routing §15 + §11.4.1 script-abort)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected files:** tests/final-verify.sh (audit B5), tests/verify-proxy.sh (audit B6)
- **Bluff audit:** docs/research/existing_test_bluffs_audit/README.md (B5, B6)
- **Future guard:** the standing assert_egress_ip invocation (live RED/GREEN at P10, §11.4.135) — SKIPS honestly until a real tunnel + VPN_EXIT_IP exists.
- **Workable item:** #49 (P8b)

Symptom & root cause (FACT — §11.4.6)

Both scripts declared VPN routing verified **PASS** when the egress IP seen THROUGH the proxy EQUALS the host's real direct IP (`host_ip == proxy_ip`). That equality is PROOF traffic was **NOT** routed through any VPN — both paths exit the same address — so the test greened the exact no-VPN configuration it claimed to forbid (the §15 bluff; `verify-proxy.sh:46`'s comment literally encoded the wrong invariant "proxy uses same IP as host"). Neither compared against an EXPECTED tunnel-exit IP. Additionally both `test_pass/test_fail` used `((PASS++))/((FAIL++))`, which returns exit 1 when the counter is 0 and aborts the script under `set -euo pipefail` (the §11.4.1 class of BUGFIX-0001/B4) — so the VPN block was unreachable until that was fixed too.

Fix (at source)

Remove the `egress==host PASS` entirely; assert the real data-plane contract via `evidence.sh`: egress through the proxy must equal the EXPECTED tunnel exit AND differ from the host IP; absent a configured `VPN_EXIT_IP` (no tunnel yet) emit an honest §11.4.3 SKIP (`operator_attended`) — never a fabricated PASS. `((PASS++))/((FAIL++))` → assignment form `PASS=$((PASS + 1))`, matching the `run-tests.sh` §11.4.1 fix.

```
- [[ "$host_ip" == "$proxy_ip" && "$host_ip" != "unknown" ]] &&
    test_pass "VPN routing verified"
+ . "$SCRIPT_DIR/lib/evidence.sh"
+ if [[ -n "${VPN_EXIT_IP:-}" ]]; then
+     assert_egress_ip "http://localhost:${HTTP_PROXY_PORT}"
+         "$VPN_EXIT_IP" "$host_ip" \
```

```
+      && test_pass "VPN routing (egress=$VPN_EXIT_IP, !=
      host)" || test_fail "...
+ else
+     ab_skip_with_reason "VPN routing (egress == expected tunnel
      exit, != host)" "operator_attended"
+ fi
```

Verification (captured, run in-session after the fix)

```
# Both run to completion vs the running proxy – connectivity PASS,
VPN SKIP:
$ bash tests/final-verify.sh -> Passed: 4 Failed: 0, exit 0
  SKIP: VPN routing (egress == expected tunnel exit, != host)
[reason: operator_attended]
$ bash tests/verify-proxy.sh -> Passed: 4 Failed: 0, exit 0

# Inverse anti-bluff proof (the §15 condition the OLD code
GREENed): set
# VPN_EXIT_IP to the host's real IP (no tunnel => egress == host)
– new code REDs:
$ VPN_EXIT_IP=<host real ip> bash tests/final-verify.sh
  FAIL: assert_egress_ip [reason: egress IP <host> == host real IP
– traffic NOT routed via VPN (§15 bluff)]
  Failed: 1, exit 1
```

The live VPN-routing GREEN (egress == real tunnel exit, != host) requires a real tunnel + VPN_EXIT_IP and is deferred to P10; until then the corrected assertion SKIPS honestly. bash -n clean both files. Reviewed independently by the conductor (§11.4.142): diffs, both runs, the inverse proof, parse — all verified against the real tree before commit.

Evidence: qa-results/p8b/.

BUGFIX-0006 — comprehensive-test.sh 100% dead ((()) set -e abort) + green-on-bluff data-plane checks + false-VPN-routing

- **Type:** Bug (anti-bluff — §11.4.1 script-abort + green-without-evidence + false-VPN-routing §15)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected files:** tests/comprehensive-test.sh (audit B1–B4, B8)
- **Bluff audit:** docs/research/existing_test_bluffs_audit/README.md (B1–B4, B8)
- **Discovered regression (§11.4.124):** #50 — the documented cache management CLI (./cache stats|clear|invalidate|trim,

docs/CACHE.md) is gone from HEAD (path collision: the ./cache CLI file vs the runtime data dir CACHE_DIR=\$PROJECT_ROOT/cache, lib/container-runtime.sh:180).

- **Workable item: #48 (P8c)**

Symptom & root cause (FACT — §11.4.6)

The whole script was **dead**: `set -euo pipefail` plus bash arithmetic counters `(( ( ... ) ) / ((PASS++)))` abort with `exit 1` the moment a counter is 0, so the suite exited before a single test ran — it could never report a failure (a §11.4.1 false-pass-by-silence, the BUGFIX-0001/0003/0005 class). Behind that abort, the checks themselves were bluffs once revived:

- **B1** declared VPN routing PASS when the egress IP through the proxy equals the host's real IP — the §15 false-VPN-routing bluff (BUGFIX-0005 class).
- **B2** "cache works" asserted nothing from Squid — no `TCP_*HIT` evidence.
- **B3** "concurrent" never checked per-request HTTP status.
- **B8** "status" never inspected a real field.

Fix (at source)

Revive + de-bluff. `(( ))` counters → assignment form `(PASS=$((PASS + 1)))`, matching the `run-tests.sh / verify-proxy.sh` §11.4.1 fixes — the 44-check suite now actually runs. Each revived check now captures real data-plane evidence:

- **B2** `assert_cache_hit` greps a real `TCP_*HIT` out of the `podman-exec'd Squid access.log` for a re-fetched object (§11.4.69 evidence path).
- **B3** asserts per-request `%{http_code} 200` across all 10 concurrent requests (10/10), not merely that curl returned.
- **B8** asserts the status command emits real fields (plain, verbose, JSON).
- **B1** `egress==host` PASS deleted → honest §11.4.3 SKIP (`operator_attended`, `VPN_EXIT_IP` unset) until the P10 real tunnel — never a fabricated PASS.
- **Cache-CLI checks** are §11.4.3 topology-aware: when the documented cache CLI is absent they SKIP citing the tracked regression #50 (the §11.4.124 investigation that found it), rather than fabricate a PASS or hard-FAIL the suite on a known-tracked gap.
- **Large-file** asserts faithful relay (`proxy_size == direct_size`); the earlier FAIL was root-caused to an EXTERNAL endpoint cap, NOT proxy truncation (evidence: `direct_size=102400 proxy_size=102400`).
- `.env` sourcing is optional — its absence is a topology SKIP, not a FAIL.

Verification (captured, run in-session after the fix — independent re-runs)

```
# Fresh clone, no .env present — runs to the summary,
deterministic across 2 runs:
$ bash tests/comprehensive-test.sh
  Tests Run: 44   Passed: 34   Failed: 0   Skipped: 10   (exit
0)   [no FAIL lines]

# Real data-plane PASSes present (not metadata):
✓ PASS: Cache HIT (Squid TCP_*HIT in access.log)
✓ PASS: Concurrent connections (10)
✓ PASS: Large file download (proxy relays faithfully)
✓ PASS: Status command reports real fields (+ verbose + JSON)

# All 10 SKIPS audited honest: 4× cache-CLI (regressed out, #50),
# 3× no-.env vars, .env-topology, VPN routing (B1), VPN container
(no-vpn mode).
```

Re-run #2 byte-identical (44/34/0/10, exit 0 — deterministic §11.4.50). Residue

- secret scan CLEAN; no ((VAR++)) remain; bash -n clean. Reviewed independently by the conductor (§11.4.142/ §11.4.134, iterate-to-GO): the lane returned with 5 hard FAILs on the first pass; the §11.4.124 investigation proved 4 were a REAL regression (#50, not a test bug → SKIP-with-reason) and 1 an external cap (faithful-relay, not truncation), and the corrected lane was re-reviewed clean before commit.

Evidence: qa-results/comprehensive/.

BUGFIX-0007 — documented cache management CLI accidentally deleted (path collision with the runtime cache/ data dir) → restored as cachectl

- **Type:** Bug (regression — documented end-user feature unusable; §11.4.124)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected files:** cachectl (restored + 3 runtime fixes), tests/run-tests.sh
 - tests/comprehensive-test.sh (cache-CLI checks → real PASS), README.md / USER_GUIDE.md / docs/CACHE.md / docs/TROUBLESHOOTING.md (./cache→./cachectl), docs/scripts/cachectl.md, tests/regression/cache_cli_present_test.sh (guard)

- **Workable item:** #50 (surfaced by BUGFIX-0006's revival of `comprehensive-test.sh`)

Symptom & root cause (FACT — §11.4.6 / §11.4.124 git-history investigation)

The documented 368-line cache management CLI (`stats|clear|invalidate|warmup|list|size|trim`, referenced in README / USER_GUIDE / docs/CACHE / docs/TROUBLESHOOTING) was **gone from HEAD** — every documented `./cache <cmd>` was a broken instruction. Git history proves an ACCIDENTAL deletion: `CACHE_DIR` has always been `$PROJECT_ROOT/cache` and `.gitignore:16` ignores `cache/` (the directory), so the tracked cache FILE and the runtime cache/ DATA directory occupy the SAME path and cannot coexist. Once the runtime materialised the directory, commit `6ec58ef` ("fix: container config for host-vpn mode") recorded the cache file as deleted in a broad `git add` — collateral to an unrelated change. The data dir `cache/{squid,streaming}` is LIVE (bind-mounted into the running proxy), so it must NOT move (§11.4.133).

Fix (at source)

Restore the CLI faithfully from `84e1754:cache` under the NON-colliding name `cachectl` (§11.4.101 safe/reversible — coexists with the gitignored `cache/ data dir`; the live data dir + running proxy untouched). The byte-restored file did NOT work under rootless Podman; three FACT-grade runtime defects fixed at source (the §11.4.108 SOURCE-restored ≠ RUNTIME-works lesson):

1. **pipefail partial-read abort** — squid's cache subdirs are owned by a remapped subuid the host cannot traverse, so `du/` `find` exit non-zero; the sourced `lib/container-runtime.sh` re-enables `set -o pipefail`, aborting every `stat/list` mid-output. Fix: `set +o pipefail` after the source.
2. **CONTAINER_RUNTIME unset** — `main()` never called `init_runtime`, so the `invalidate squid-flush` case aborted under `set -u`. Fix: `init_runtime + detect_container_runtime fallback + export` in `main()`.
3. **((removed++)) set -e abort in trim** — post-increment returns exit 1 when `removed=0`. Fix: `removed=$((removed + 1))` (the §11.4.1 class).

Docs synced (`./cache <cmd>` → `./cachectl <cmd>`, 28 replacements; the `CACHE_DIR=./cache data-dir` value correctly untouched). NB: the documented command name changes `cache`→`cachectl`; the operator may later prefer a different collision resolution (e.g. relocating the data dir) — the capability is restored now and the name choice is reversible.

Verification (captured, re-run independently by the conductor §11.4.142)

```
# cachectl works LIVE against the rootless-podman cache (read-only):
$ ./cachectl stats|size|list -> exit 0, real figures (Cache Directory, 4.0K, Files: 2, full tree)
# §11.4.135 guard (RED_MODE polarity §11.4.115):
$ tests/regression/cache_cli_present_test.sh -> GREEN
PASS (dispatches all 7 subcommands)
$ RED_MODE=1 tests/regression/cache_cli_present_test.sh -> RED
PASS (reproduces "CLI absent → unusable")
# §1.1 mutation (conductor's own): drop `trim` from dispatch -> guard FAILs
# "missing-subcommands:[trim]" (bash -n still clean = assertion, not parse error);
# restored byte-identical md5 7e935b1deb4fed1435f6b48274844c2f (before==after).
# Suites (conductor re-run):
$ bash tests/comprehensive-test.sh -> 44 run / 38 pass / 0 fail / 6 skip (the 4 cache-CLI checks now PASS)
$ bash tests/run-tests.sh -> 45 run / 39 pass / 0 fail / 6 skip, exit 0 (incl. BUGFIX-CACHECLI GREEN+RED)
```

Honest gap (§11.4.6): the invalidate squid-flush WIRING is proven (CONTAINER_RUNTIME=podman, is_container_running proxy-squid=TRUE, podman branch selected) but the live squid -k rotate was not executed end-to-end (target-safety §11.4.133 — not flushing the live proxy's cache); destructive subcommands were proven on a throwaway dir. warmup remains the original documented placeholder.

Evidence: qa-results/cachectl/, qa-results/regression/cachecli/.

BUGFIX-0008 — un-audited mutation window: control-API mutate+audit were two separate store calls (audit-fail left the mutation persisted)

- **Type:** Bug (latent data-integrity defect — transactional atomicity; P6 WARNING-4)
- **Status:** Fixed
- **Date:** 2026-07-01
- **Affected files:** control-plane/internal/store/store.go (WithTx), internal/store/postgres.go (dbExecutor seam + real tx), internal/api/handlers.go (mutateWithAudit), 2 test fakes, internal/api/atomicity_test.go (§11.4.135 guard), internal/store/postgres_integration_test.go (real-PG proof)

- **Workable item:** #52 (P6 review WARNING-4 follow-up)

Symptom & root cause (FACT — §11.4.102/ §11.4.124 investigation)

All 9 mutating control-API handlers did TWO separate store calls — Upsert*/Delete* then a distinct AppendAudit — with no transaction spanning them. Proven on the pre-fix artifact (RED_MODE=0 FAILED): when AppendAudit fails, the mutation is already durable and the handler returns HTTP 500 → an **un-audited mutation persists** (a silent integrity hole: the audit trail no longer reflects the data).

Fix (at source, §11.4.150 pgx-tx research)

One store-interface method WithTx(ctx, func(tx Queries) error) error (chosen over 9 per-op *WithAudit methods — 1 vs 9, audit SQL in one place). Real impl begins a *sql.Tx (a dbExecutor seam lets both *sql.DB and *sql.Tx satisfy the query path with zero SQL duplication), commits on nil / rolls back on error; the fake snapshot-restores. The handler seam mutateWithAudit runs mutate(tx) + AppendAudit inside ONE WithTx — the mutation error passes through UNWRAPPED (so errors.Is(..., ErrNotFound)→404 is preserved), the audit error is wrapped (→500 + rollback). The store owns begin/commit/rollback; the handler never orchestrates a transaction it cannot roll back.

Verification (captured, re-run independently by the conductor §11.4.142)

```
# §11.4.135 guard atomicity_test.go (RED_MODE polarity, 3 entity
types):
# RED_MODE=0 GREEN – audit-fail rolls back the mutation (0
profiles / row restored / 0 targets)
# RED_MODE=1 reproduces the defect on the 2-call pre-fix path
# Conductor §1.1 mutation: swallow the audit error in the WithTx
seam ->
# atomicity_test FAILs with real assertions (audit-fail returns
200/204 not 500,
# across all 3 subtests) -> byte-identical restore md5
a0fb9b65dac430e641130200bcb802d4.
$ go test -race -count=3 -short ./internal/... -> all 8
packages ok, no data races (§11.4.50)
$ go test -run TestIntegration_WithTxAtomicAuditAndMutation ./
internal/store/
-> PASS 4.76s (real postgres:16-alpine: commit persists both;
rollback leaves NEITHER)
$ go build/vet ./... exit 0; gofmt -l . empty
```

Honest gaps (§11.4.6): the `-count=3` determinism sweep is the logic layer (the real-PG integration is proven once — booting N postgres containers is wall-time non-deterministic by nature, §12.6); the fake `WithTx` is single-process snapshot/restore (real atomicity proven separately against live PG); nested `WithTx` is unsupported (documented; handlers never nest). Operator PG (`lava-postgres-thinker`) never touched — the integration test boots a throwaway `hp-it-pg-<port>` (`--rm, t.Cleanup`).

Evidence: pasted `-race/integration` output; test at `internal/api/atomicity_test.go`.